

Task 7: SQL Queries for Data Extraction

Objective:

To learn SQL data extraction using SELECT queries, JOIN operations, and GROUP BY aggregation.

Tools Used:

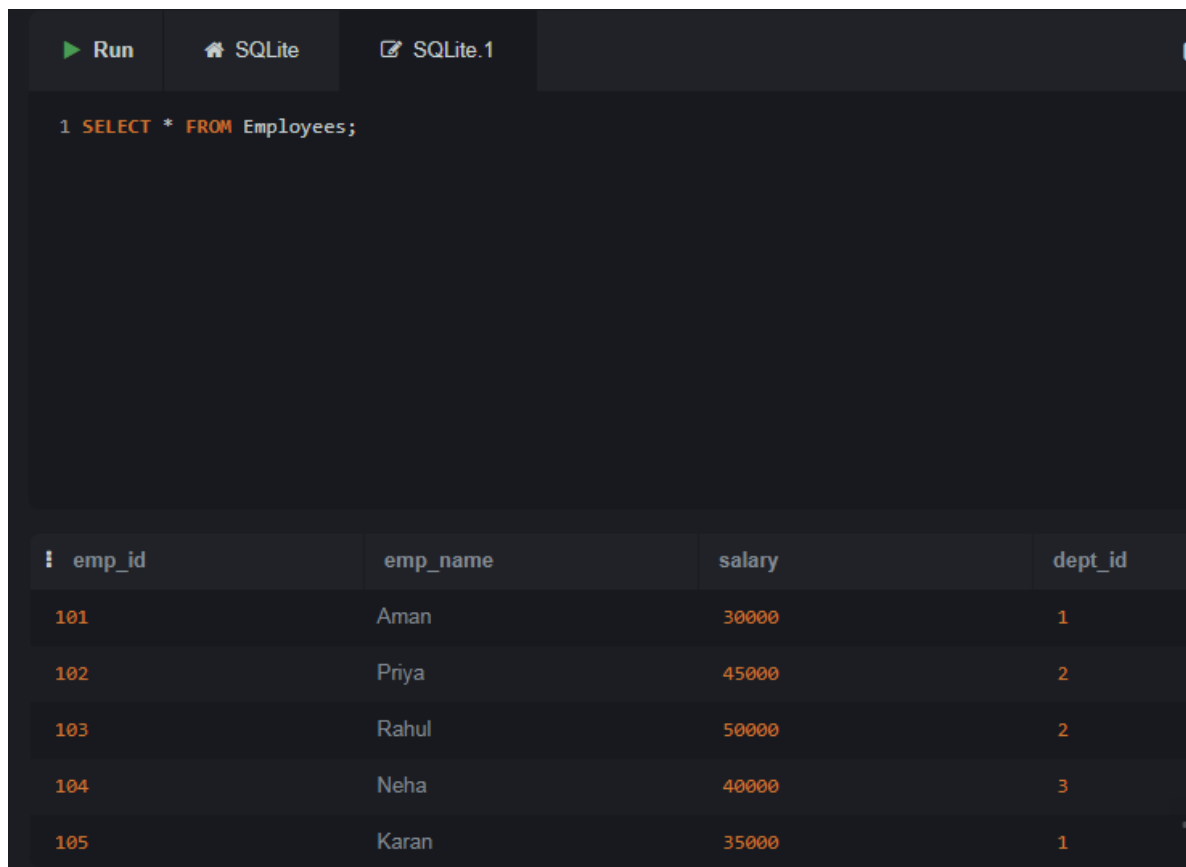
SQLite Online

Database Tables:

1. Departments
2. Employees

Queries Performed:

1. SELECT Query
2. WHERE Clause
3. INNER JOIN
4. GROUP BY with AVG()
5. COUNT()
6. Department



The screenshot shows the SQLite Online interface. At the top, there are tabs for 'Run', 'SQLite', and 'SQLite.1'. Below the tabs, a SQL query is entered in a text area: `1 SELECT * FROM Employees;`. Below the query area, the results are displayed in a table with 4 columns: `emp_id`, `emp_name`, `salary`, and `dept_id`. The results table contains 5 rows of data.

emp_id	emp_name	salary	dept_id
101	Aman	30000	1
102	Priya	45000	2
103	Rahul	50000	2
104	Neha	40000	3
105	Karan	35000	1

▶ Run

🏠 SQLite

📄 SQLite.1

```
1 SELECT * FROM Employees
2 WHERE salary > 40000;
```

emp_id	emp_name	salary	dept_id
102	Priya	45000	2
103	Rahul	50000	2

Run	SQLite	SQLite.1
<pre> 1 SELECT e.emp_name, 2 d.dept_name, 3 e.salary 4 FROM Employees e 5 INNER JOIN Departments d 6 ON e.dept_id = d.dept_id; </pre>		
emp_name	dept_name	salary
Aman	HR	30000
Priya	IT	45000
Rahul	IT	50000
Neha	Finance	40000
Karan	HR	35000

Run

SQLite

SQLite.1

```
1 SELECT * FROM Employees
2 WHERE salary > 40000;
```

emp_id	emp_name	salary	dept_id
102	Priya	45000	2
103	Rahul	50000	2

Run	SQLite	SQLite.1
<pre> 1 SELECT dept_id, 2 COUNT(*) AS total_employees 3 FROM Employees 4 GROUP BY dept_id; </pre>		
dept_id	total_employees	
1	2	
2	2	
3	1	

Run	SQLite	SQLite.1
<pre> 1 SELECT * FROM Departments; </pre>		
dept_id	dept_name	
1	HR	
2	IT	
3	Finance	

Results:
(Add all screenshots here)

Conclusion:
Successfully extracted and analyzed data from a relational database using SQL queries.